Can Climate Policy Build Peace?

Revenue Earmarking, Policy Bundling, and Public Support for Carbon Taxation in Colombia

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Abstract

1 Theoretical Puzzle and Research Questions

Climate mitigation and peacebuilding represent two major policy challenges that require sustained political support to overcome collective action problems, distribute costs, and deliver benefits that may materialize only over time. Yet the political constituencies supporting these objectives do not necessarily overlap. Climate mitigation policies, particularly carbon taxation, impose relatively immediate and visible economic costs while generating environmental benefits that are often diffuse, uncertain, and temporally distant. Peacebuilding policies confront a related but distinct challenge: they require sustained public investment in conflict recovery, reconciliation, and institutional transformation, while directing resources toward objectives and beneficiaries that may be politically contested. In societies emerging from armed conflict, disagreements over responsibility for past violence, the distribution of public resources, and the appropriate beneficiaries of peacebuilding may persist long after formal peace agreements are signed.

Policy bundling offers a potential strategy for overcoming these political constraints by linking objectives that appeal to different constituencies within a single policy proposal. Rather than relying exclusively on environmental motivations to generate support for a carbon tax, policymakers may allocate a portion of its revenues to peacebuilding programs that citizens value for other reasons. In this account, bundling may broaden the political appeal of climate policy by incorporating benefits associated with conflict recovery, reparative justice, or rural development. Individuals who would otherwise be reluctant to support a costly environmental policy may become more supportive when that same policy also advances a peace-related objective.

Importantly, the potential political advantage of bundling does not require citizens to share the same motivations for supporting a policy. Environmentally motivated constituencies may support a carbon tax because of its contribution to climate mitigation or environmental protection, while peace-oriented constituencies may support the same proposal because it finances programs they consider socially or politically valuable. A climate–peace bundle could therefore create a common basis for policy support among groups with different priorities. From this perspective, the political value of bundling lies not simply in increasing the number of benefits associated with a policy, but in its potential to connect constituencies that would not otherwise support the same

intervention. However, combining climate mitigation and peacebuilding does not necessarily produce a broader coalition. Carbon-tax revenues constitute a finite pool of resources, meaning that allocating a greater share to peacebuilding reduces the share available for environmental expenditures. Individuals who prioritize environmental protection may therefore oppose peace earmarking if they perceive it as diverting resources away from the original objectives of the policy. At the same time, peacebuilding is itself politically contested. Citizens may disagree over whether public resources should finance victim reparations, ex-combatant reintegration, illicit-crop substitution, or rural infrastructure. The introduction of peace-related benefits may consequently attract some constituencies while alienating others.

These competing responses create an important distinction between increasing average policy support and broadening the constituencies willing to support a policy. A climate–peace bundle may attract individuals who were previously skeptical of climate policy without substantially reducing support among existing environmental constituencies. Alternatively, gains among peace-oriented respondents may be offset by losses among those who oppose the peacebuilding component or the reallocation of environmental revenues. Bundling may therefore generate coalition expansion, coalition substitution, or even a contraction in the potential support base. Similar average effects may conceal substantially different patterns of political support across constituencies. Moreover, the political consequences of bundling may depend on how the policy combines its objectives. The allocation of revenues between environmental and peacebuilding purposes determines the relative priority given to each objective, while the specific peacebuilding program identifies the benefits and beneficiaries associated with the policy.

These choices may shape which constituencies find the bundle attractive and which perceive it as politically or distributively undesirable. Household costs introduce an additional constraint: even when citizens value both climate mitigation and peacebuilding, the expected financial burden of the policy may limit their willingness to support it. The presentation and endorsement of the policy may also influence its political appeal. Describing an otherwise identical instrument as a carbon tax, climate contribution, or climate investment may alter the relative salience of its costs and anticipated benefits. Similarly, endorsement by the national government, an international organization, an environmental organization, or a group of universities may provide different signals about the credibility or desirability of the proposal. These possibilities suggest that the political effects of bundling may depend not only on substantive policy characteristics, but also on how the proposal is communicated and by whom it is endorsed.

These competing mechanisms generate the central theoretical puzzle of this study: *Can linking climate mitigation to peacebuilding broaden the political constituencies supporting a costly environmental policy, or does combining these objectives instead generate offsetting political gains and losses?* More specifically, under what conditions does climate–peace bundling increase support, which constituencies respond to the introduction of peacebuilding, and how do the composition and costs of the bundle shape its political appeal? Colombia provides a substantively important setting in which to examine these questions. Carbon taxation creates an opportunity to investigate public preferences over the allocation of climate-policy revenues, while the country's experience of armed conflict and peacebuilding makes peace-related expenditures politically salient and potentially contested. By examining whether citizens respond differently to environmental-only earmarking and alternative climate–peace bundles, this study investigates the potential for cross-issue policy bundling to broaden support for climate action in a post-conflict setting.

The analysis is organized around the following research questions. I distinguish directional hypotheses about the expected effects from the numerical effect-size assumptions used to plan statistical power. The latter are illustrative calibration inputs, not estimates of Colombian public opinion:

- **RQ1. Average effect of bundling:** *Does earmarking a portion of carbon-tax revenue for peacebuilding increase public support for climate policy relative to allocating all revenues to environmental purposes?*
- **RQ2. Coalition formation:** *Does climate–peace bundling attract constituencies otherwise reluctant to support climate policy without alienating existing supporters, or does it generate offsetting gains and losses across constituencies?*
- **RQ3. Characteristics of the climate–peace policy bundle:**
  - **RQ3a. Revenue composition:** *How does the allocation of carbon-tax revenue between environmental and peacebuilding objectives shape public support?*
  - **RQ3b. Peacebuilding expenditure:** *Does the type of peacebuilding expenditure condition the political appeal of climate–peace policy bundles?*
  - **RQ3c. Household costs:** *How do household costs affect support for carbon-tax policies?*

As secondary extensions, the study may also examine whether policy naming and institutional endorsements influence support for carbon-tax proposals and whether these cues condition the effects of climate–peace bundling. These analyses are intended to complement, rather than replace, the primary focus on revenue allocation and the political constituencies supporting climate policy.

2 Testable implications (attributes of the conjoint)

2.1 Policy-bundling appeal and counterbalancing concerns

Carbon taxes impose relatively immediate and visible costs on households, while their environmental benefits may be delayed, diffuse, and difficult for individuals to attribute directly to the policy. Earmarking carbon-tax revenues for identifiable public expenditures may help counterbalance these perceived costs by making the benefits of climate policy more concrete. In particular, allocating a portion of revenues to peacebuilding may increase the appeal of a carbon tax among individuals who value conflict recovery, reparative justice, or development in historically conflict-affected communities, even if climate mitigation is not their primary political priority. The central theoretical proposition is that combining climate mitigation and peacebuilding within a single policy may broaden the set of objectives served by a carbon tax and, consequently, the constituencies willing to support it. However, bundling does not necessarily increase support. Carbon-tax revenues constitute a finite pool of resources: allocating a larger share to peacebuilding necessarily reduces the share available for environmental purposes. Individuals who strongly prioritize environmental spending may therefore perceive peace earmarking as a diversion from the original objectives of climate policy. The net effect of bundling depends on whether the additional political support generated by peace-related expenditures outweighs the support potentially lost through this reallocation.

This logic motivates the primary comparison in RQ1. The conjoint experiment distinguishes proposals that allocate all carbon-tax revenues to environmental purposes from proposals that allocate a positive share to peacebuilding. The primary estimand is the equally weighted average effect of the three bundled configurations (25%, 50%, and 75% peace allocations), relative to the exclusively environmental allocation. This contrast captures whether introducing peacebuilding into the allocation of carbon-tax revenues increases or decreases policy support on average, across the positive peace shares and eligible peacebuilding programs included in the experiment. Importantly, this estimand identifies the effect of reallocating revenue from exclusively environmental purposes toward a climate–peace bundle. It does not identify the effect of adding peacebuilding benefits while holding environmental expenditure constant. The environmental and peacebuilding components are jointly determined by the revenue-allocation attribute.

The same design also permits a more specific test of how the composition of the bundle shapes support (RQ3a). Conditional on a positive peace allocation, the analysis compares the 25%, 50%, and 75% peace shares against one another. Pairwise contrasts between these configurations identify whether environment-prioritizing, balanced, or peace-prioritizing bundles are more politically attractive. These comparisons are distinct from the primary bundling estimand: RQ1 evaluates the introduction of peacebuilding relative to an all-environment policy, whereas RQ3a evaluates trade-offs among alternative climate–peace bundles. Taken together, these comparisons distinguish whether bundling is politically advantageous on average from whether its effectiveness depends on the relative prioritization of environmental and peacebuilding objectives. The underlying mechanisms may involve benefit salience, distributive preferences, or perceived policy effectiveness; the conjoint estimates their combined implications for stated policy support rather than isolating each mechanism independently.

Hypothesis H1: Average effect of peace earmarking (RQ1). I expect that, on average, a carbon-tax proposal allocating a positive share of revenues to peacebuilding will be less likely to be selected than a proposal allocating all revenues to environmental purposes. This expectation reflects the possibility that redirecting environmental-tax revenues toward a politically contested objective generates opposition that exceeds the additional support attracted by peace-related expenditures. The direction is not mechanically implied by bundling: the average effect may be positive if the appeal of peacebuilding outweighs these costs.

Hypothesis H3a: Revenue-allocation gradient (RQ3a). Among proposals that include peacebuilding, I expect the probability of selection to decline as a larger share of carbon-tax revenues is allocated to peace rather than environmental expenditures. In particular, I expect 25% peace earmarking to be preferred, on average, to 50%, and 50% to 75%. This expectation is consistent with possible revenue-diversion aversion or perceived policy incoherence, although the conjoint cannot independently identify these mechanisms.

2.2 Political contestation and coalition trade-offs

The political consequences of climate–peace bundling are unlikely to be uniform across the population. Individuals differ in their priorities concerning climate mitigation, peacebuilding, redistribution, and the role of government. A policy bundle that combines multiple objectives may therefore attract support from some constituencies while simultaneously generating opposition among others. This possibility is central to the distinction between coalition expansion and coalition substitution.

Bundling may expand the potential support base for climate policy if peace-related benefits attract individuals who would otherwise be reluctant to support a carbon tax, without substantially alienating existing climate-policy supporters. Alternatively, bundling may produce coalition substitution if gains among peace-oriented constituencies are offset by losses among environmentally oriented constituencies. A contraction in support is also possible if the introduction of peacebuilding generates more opposition than additional support. The experiment examines these possibilities by distinguishing four baseline constituencies according to respondents' attitudes toward climate policy and peacebuilding:

- Pro-climate and pro-peace;
- Pro-climate and peace-skeptical;
- Climate-skeptical and pro-peace; and
- Skeptical of both objectives.

These constituencies are defined using respondents' measured attitudes rather than randomized treatment assignments. To avoid conditioning on attitudes potentially affected by exposure to the conjoint, the primary constituency analysis uses the survey-order arm in which climate and peacebuilding attitudes are measured before the conjoint. RQ2 estimates the average bundling effect separately for each baseline constituency. For each group, the estimand compares the equally weighted average support for the three positive-peace allocations with support for the exclusively environmental allocation. Differences between these group-specific effects reveal whether the same policy bundle generates systematically different responses across constituencies.

For example, a positive bundling effect among climate-skeptical but peace-supportive respondents would be consistent with the possibility that peace-related benefits recruit individuals who are not primarily motivated by climate mitigation. A negative effect among pro-climate but peace-skeptical respondents would be consistent with the possibility that bundling alienates some existing climate-policy supporters. The relative magnitude and direction of these effects provide evidence about the political trade-offs created by combining climate and peacebuilding objectives. Examining these heterogeneous effects is important because the political value of bundling cannot be understood through average policy support alone. Two policy bundles may generate similar average effects while producing very different patterns of support across constituencies. A bundle that attracts previously climate-skeptical respondents without alienating existing climate-policy supporters has different implications for coalition formation than one that generates offsetting gains and losses across groups.

Distinguishing these patterns helps explain whether linking climate policy to peacebuilding can broaden its potential political support base or instead reconfigure existing political divisions. This is particularly relevant when climate and peacebuilding objectives appeal to constituencies with different priorities, interests, and distributive preferences. Nevertheless, heterogeneous conjoint effects provide evidence of potential coalition recruitment and alienation rather than direct evidence of organized political coalition formation. Forced-choice outcomes measure relative preferences between policy proposals, not absolute willingness to support a policy. Consequently, claims about coalition expansion must consider baseline support, constituency sizes, and the limitations of stated-preference measures.

This framework links the average bundling effect in RQ1 to the distribution of its political consequences in RQ2. It also provides a basis for interpreting the program-specific comparisons in RQ3b: different peacebuilding objectives may alter the constituencies attracted to or alienated by a climate–peace bundle.

Hypothesis H2: Constituency-specific responses (RQ2). I expect the effect of peace earmarking to differ according to respondents' prior attitudes toward climate policy and peacebuilding. Specifically, I expect the most clearly positive response among respondents who are skeptical of climate policy but supportive of peacebuilding, and a negative response among those supportive of climate policy but skeptical of peacebuilding. My focal between-constituency hypothesis is therefore that the bundling effect is more positive among the climate-skeptical/pro-peace constituency than among the pro-climate/peace-skeptical constituency. For respondents supportive of both objectives, the direction is ambiguous because peace benefits must be weighed against forgone environmental spending. Among respondents skeptical of both, I expect a nonpositive effect, although its magnitude is uncertain. Because constituency membership is not randomized, it does not identify the causal effect of holding a particular attitude. Nor does a positive within-group forced-choice effect alone demonstrate that the total coalition supporting carbon taxation has expanded; this is essentially a heterogeneous effect exercise.

2.3 Program-specific valuation and credibility

Peacebuilding is not a unitary policy objective. Different peacebuilding expenditures may evoke distinct political commitments, distributive preferences, and expectations about government implementation. Consequently, the political appeal of combining climate policy with peacebuilding may depend not only on the share of revenues allocated to peace-related objectives, but also on the specific programs being financed. The experiment considers four peacebuilding expenditures: victim reparations, ex-combatant reintegration, illicit-crop substitution, and rural infrastructure. These programs represent different dimensions of post-conflict recovery and may appeal to different constituencies. Victim reparations may be associated with recognition and reparative justice, while ex-combatant reintegration may evoke competing views about reconciliation, accountability, and the distribution of public resources. Illicit-crop substitution may be valued for its potential contribution to alternative rural livelihoods, whereas rural infrastructure may appeal through broader development benefits and improvements in public-service provision.

These distinctions are theoretically important because the effectiveness of climate–peace bundling may depend on the political constituencies and beneficiaries associated with each expenditure. A program that appeals to individuals otherwise reluctant to support climate policy could help broaden its support base. Conversely, a program whose beneficiaries are politically contested could generate opposition, including among respondents who would otherwise favor the environmental objectives of the carbon tax. Thus, the choice of peacebuilding expenditure may help explain why some climate–peace bundles attract support while others generate resistance. Program-specific preferences may also reflect differences in perceived credibility and implementation capacity. Respondents may be more willing to support a carbon tax when they believe that earmarked revenues will finance tangible and achievable public benefits. Other programs may be viewed as less credible because of distrust in implementing institutions, uncertainty about beneficiaries, or skepticism about whether promised expenditures will materialize. These are plausible interpretations of program-specific effects, but they cannot be separately identified from the program contrasts alone.

RQ3b evaluates whether the type of peacebuilding expenditure conditions the political appeal of climate–peace bundles. The peacebuilding-program attribute is randomized only when the proposal allocates a positive share of revenue to peace. The analysis therefore estimates conditional program effects, comparing the average marginal component effects (AMCEs) of alternative programs within bundled proposals. Pairwise comparisons are averaged over a common distribution of positive peace allocations and other randomized policy characteristics. These comparisons identify whether different peacebuilding objectives generate different levels of support for the same type of climate policy. They also distinguish the role of program choice from the relative allocation of revenues between environmental and peacebuilding purposes. However, a higher AMCE for one peacebuilding program does not, by itself, establish that the program expands the climate-policy coalition. That interpretation requires examining which constituencies respond to the program and whether their support increases or decreases.

The theoretical contribution of RQ3b is therefore not simply to rank peacebuilding programs by popularity. Rather, it is to examine whether the political effectiveness of linking climate policy to peacebuilding depends on the particular objectives and beneficiaries incorporated into the bundle.

Hypothesis H3b: Peacebuilding expenditure (RQ3b). Conditional on positive peace earmarking, I expect proposals financing rural infrastructure to receive the greatest support, followed by victim reparations, illicit-crop substitution, and ex-combatant reintegration. I expect rural infrastructure to appeal through benefits that may extend beyond narrowly defined beneficiaries, whereas reintegration may generate greater contestation over deservingness and the allocation of public resources. These expectations are theoretically motivated rather than drawn from directly comparable estimates.

2.4 Cost sensitivity

The political appeal of carbon-tax policies depends not only on the benefits they promise but also on the costs households expect to bear. Even respondents who value environmental protection or peacebuilding may be unwilling to support a proposal when its perceived financial burden becomes sufficiently high. Consequently, the ability of peace-related benefits to increase support for climate policy may be constrained by household costs. The conjoint experiment varies the expected monthly household cost of each carbon-tax proposal across four levels: COP 5,000, COP 15,000, COP 30,000, and COP 60,000. Explicitly presenting these costs helps ensure that respondents evaluate proposals against a common set of experimentally specified financial trade-offs, rather than inferring different costs from the programs or revenue allocations described. RQ3c estimates the effect of household costs on policy support through pairwise comparisons of cost-level AMCEs, using COP 5,000 as the reference category. These contrasts identify how the probability of supporting or selecting a proposal changes as its expected monthly household cost increases, averaging over the prespecified distribution of other randomized policy attributes.

Because the cost attribute is randomized independently of revenue allocation, its effects can be distinguished from the political appeal of the expenditures financed by the carbon tax. This is important for interpreting the bundling estimand: any estimated average advantage of climate–peace bundles is evaluated over the same experimentally assigned distribution of household costs, rather than reflecting systematically cheaper or more expensive bundled proposals. The primary cost analysis focuses on differences in support across the four randomized cost levels. These comparisons do not require assuming that cost effects are linear. Although the design may also permit

an examination of whether the political appeal of bundling changes as household costs rise, such interactions would constitute a separate analysis from the primary pairwise cost contrasts. Overall, the cost attribute situates the evaluation of climate–peace bundling within a realistic distributive trade-off: respondents must consider not only which objectives a carbon tax finances, but also the financial burden associated with the policy.

Hypothesis H3c: Household costs (RQ3c). I expect the probability of selecting a carbon-tax proposal to decline as its stated monthly household cost increases. This expectation reflects the immediate and tangible financial burden imposed on households, particularly when the anticipated environmental or peace benefits are diffuse, uncertain, and may materialize only over time. As a result, respondents may be less willing to bear higher costs for benefits that are difficult to observe or directly attribute to the policy.

2.5 Policy naming and the framing of climate costs

Public support for climate policy may depend not only on its substantive design but also on how the policy is presented to citizens. Even when two proposals impose identical household costs and allocate revenues to the same objectives, the terminology used to describe them may influence how respondents interpret their economic burden, intended benefits, and broader political purpose. The experiment considers three alternative names for the same policy instrument: "Impuesto al carbono" (carbon tax), "Contribución climática" (climate contribution), and "Inversión climática" (climate investment). These labels emphasize different aspects of the policy. The term "carbon tax" explicitly identifies the instrument as taxation and may make its financial burden particularly salient. By contrast, "climate contribution" may evoke collective responsibility and shared participation in addressing climate change, while "climate investment" may direct attention toward the future benefits generated by public expenditure.

These distinctions may be especially relevant for climate–peace bundling. One potential advantage of earmarking carbon-tax revenues for peacebuilding is that it makes the benefits of climate policy more concrete and connects the policy to objectives beyond emissions reduction. A policy framed as an investment or contribution may reinforce this interpretation, whereas a tax label may direct attention toward the immediate financial burden. Consequently, the political appeal of bundling may depend partly on whether the policy is presented primarily as a cost or as a collective investment in public benefits. The primary expectation is that the labels "climate contribution" and "climate investment" may generate greater support than "carbon tax," holding household costs, revenue allocations, and other policy characteristics constant. However, the direction of these effects is not unambiguous. Respondents may perceive alternative labels as euphemistic or misleading if they recognize that the underlying instrument remains a tax. Such reactions could reduce trust and offset any positive framing effects.

In the expanded design, the policy-name attribute is randomized independently across conjoint profiles. Its effects could be estimated through pairwise comparisons of average marginal component effects (AMCEs), using "carbon tax" as the reference category. These comparisons would identify whether alternative labels change stated policy support when the substantive characteristics of the proposal remain experimentally comparable. A secondary analysis could examine whether the effect of peace earmarking varies across policy labels. Such interactions would help assess whether the political benefits of climate–peace bundling are reinforced or weakened by the way the policy is presented. Nevertheless, these effects would identify responses to policy labels rather

than separately isolating the psychological mechanisms of cost salience, collective responsibility, or perceived investment benefits.

Secondary hypothesis H4: Policy naming. Holding the substantive attributes constant, I expect proposals described as a "climate contribution" or "climate investment" to be selected more often than proposals described as a "carbon tax," with the contribution label producing the larger effect. This expectation reflects how policy labels shape perceptions of government intervention and individual financial obligations. While the term "tax" may emphasize an additional financial burden and trigger resistance to taxation, "contribution" may evoke a sense of shared responsibility toward a collective goal, and "investment" may emphasize the potential long-term returns of climate action. I expect the contribution label to be particularly effective because it frames the payment as an act of collective participation rather than a compulsory financial obligation, potentially making the policy more acceptable even when its actual costs and benefits remain unchanged.

2.6 Institutional endorsement and source credibility

Citizens' evaluations of climate policy may also depend on the institutions associated with a proposal. Public policies often involve uncertainty about implementation, the distribution of benefits, and whether governments will fulfill their commitments. Under these conditions, institutional endorsements may provide cues about the credibility, technical quality, or political orientation of a proposed policy. This consideration may be particularly important for climate–peace bundling, which combines two policy objectives that require substantial government coordination and public expenditure. Respondents may support the idea of allocating carbon-tax revenues to environmental protection and peacebuilding while remaining skeptical about whether the promised benefits will actually be delivered. The institution endorsing the proposal may therefore influence how respondents evaluate the credibility and desirability of the bundle.

The experiment considers four potential endorsing entities: the national government, the United Nations, a Colombian environmental organization, and a group of Colombian universities. These entities may evoke different forms of legitimacy and expertise. The national government may be associated with formal implementation authority but also with political polarization or distrust. The United Nations may signal international oversight or expertise in peacebuilding and environmental governance. A Colombian environmental organization may provide a cue of environmental commitment, while Colombian universities may signal technical knowledge and relative independence from partisan politics. The expected effects are theoretically ambiguous. Endorsements from organizations perceived as credible or technically competent may increase support relative to endorsement by the national government, particularly among respondents with low trust in public institutions. However, international organizations may also provoke skepticism among respondents who oppose external involvement in domestic policymaking. Similarly, environmental organizations and universities may be trusted by some constituencies but viewed as politically or socially distant by others.

Institutional endorsements may also condition the political effects of bundling. For example, an endorsement associated with peacebuilding expertise may increase confidence that peace-related revenues will be used effectively, while an environmental endorsement may reassure respondents concerned about diverting resources away from environmental objectives. In this sense, institutional cues could influence whether the introduction of peacebuilding benefits attracts additional supporters or generates opposition. In the expanded design, the endorsing-entity attribute is ran-

domized independently across conjoint profiles. Pairwise AMCE comparisons, using the national government as the reference category, would estimate how different endorsements affect policy support. Additional interactions with revenue allocation could assess whether the average bundling effect differs according to the endorsing institution. These comparisons would identify the causal effects of experimentally assigned endorsement cues on stated preferences. They would not, on their own, establish whether any observed effects operate through institutional trust, perceived implementation capacity, political identification, or other interpretations of the endorsing entity.

Secondary hypothesis H5: Institutional endorsement. I expect endorsement by Colombian universities to increase policy selection relative to endorsement by the United Nations, followed by a smaller positive effect for a Colombian environmental organization, while endorsement by the national government is expected to be less attractive on average. This ordering reflects expectations about technical credibility, institutional trust, and political contestation. Colombian universities may be perceived as relatively independent sources of scientific expertise and policy knowledge, lending credibility to the environmental objectives and expected benefits of carbon taxation. Their domestic character may also make their endorsement more relevant and relatable to Colombian respondents than that of an international organization such as the United Nations. Colombian environmental organizations may similarly signal expertise and commitment to climate protection, although their endorsement could be less persuasive among respondents who perceive environmental advocacy as politically or ideologically motivated. In contrast, endorsement by the national government may activate existing political loyalties and distrust, shifting attention away from the substantive merits of the proposal and toward evaluations of the incumbent administration. I therefore expect endorsements from institutions perceived as technically credible and relatively independent of partisan politics to be more effective in increasing policy selection.

3 Conjoint Experimental Design

3.1 Experimental overview

To examine whether linking climate mitigation to peacebuilding increases public support for carbon taxation, this study employs a randomized conjoint experiment among adults in Colombia. Respondents evaluate hypothetical carbon-tax proposals that vary in their revenue allocations, peacebuilding expenditures, and expected household costs. The expanded design also varies policy naming and institutional endorsements, which are analyzed as secondary attributes; a reduced design without these two attributes is evaluated as a sensitivity scenario in the power analysis. The experiment is designed to distinguish whether introducing peacebuilding into climate-policy revenue allocation increases average support, whether its effects differ across political constituencies, and which characteristics of the resulting policy bundles shape their political appeal.

Figure 1 summarizes the experimental structure, including the respondent-level survey-order randomization and the attributes randomized within the conjoint. Table 1 connects the research questions to the corresponding experimental attributes, econometric estimands, and substantive interpretations. The preliminary recruitment target is 3,000 completed and usable surveys, subject to feasibility and recruitment costs; 2,000 completed surveys remains an important benchmark in the power analysis. Recruitment is planned through Facebook advertisements targeting adults in Colombia. As illustrated in Figure 1, each respondent completes six paired-profile conjoint tasks. In each task, respondents compare two hypothetical carbon-tax proposals and indicate which one

they prefer. At $N = 3{,}000$, this structure yields 18,000 paired-choice tasks and 36,000 policy-profile evaluations, assuming all six tasks are completed. At the earlier $N = 2{,}000$ benchmark, the corresponding figures are 12,000 tasks and 24,000 profiles. Randomizing policy characteristics across profiles allows the study to identify their effects on stated preferences while averaging over the distribution of other experimentally assigned attributes.

3.2 Policy attributes and randomization

The central experimental attribute is the allocation of carbon-tax revenues between environmental and peacebuilding objectives. As shown in Table 1, the design includes an exclusively environmental allocation and three alternatives that earmark a positive share of revenues for peacebuilding. This variation serves two distinct purposes. First, comparing the positive-peace allocations with the all-environment baseline identifies the average effect of introducing peacebuilding into the policy (RQ1). Second, comparisons among the positive-peace allocations identify whether support changes as a larger share of revenues is directed toward peacebuilding rather than environmental purposes (RQ3a). These comparisons are theoretically distinct. The first evaluates whether bundling climate and peacebuilding objectives is politically advantageous relative to exclusively environmental earmarking. The second evaluates how the composition of a climate–peace bundle affects support once peacebuilding is included. The conjoint also varies the peacebuilding program financed by the proposal. As summarized in Table 1, the four programs are victim reparations, ex-combatant reintegration, illicit-crop substitution, and rural infrastructure.

Importantly, this attribute is randomized conditionally on the revenue allocation. When a proposal allocates no revenue to peacebuilding, no peacebuilding program is assigned. When the peace share is positive, one of the four programs is randomly selected. This conditional structure, illustrated in Figure 1, ensures that the hypothetical proposals remain internally consistent. It also determines the interpretation of RQ3b: the program-specific estimands compare alternative peacebuilding expenditures within bundled policies, rather than comparing programs across all possible revenue allocations. A third attribute varies the expected monthly household cost of the policy across four levels, ranging from COP 5,000 to COP 60,000. As indicated in Table 1, RQ3c examines how changes in these costs affect support for carbon-tax proposals. Household cost is randomized independently of revenue allocation and peacebuilding program, allowing the experiment to distinguish preferences over policy benefits from responses to its financial burden.

Finally, Figure 1 includes two secondary attributes concerning the presentation of the policy. The first varies whether the proposal is described as an *impuesto al carbono*, *contribución climática*, or *inversión climática*. The second varies whether the proposal is endorsed by the national government, the United Nations, a Colombian environmental organization, or a group of Colombian universities. The current power-analysis baseline includes these attributes as independently randomized components of the expanded conjoint; their inclusion in the final survey instrument remains subject to final design decisions. They are not part of the primary estimands reported in Table 1.

3.3 Outcome measurement

As illustrated in Figure 1, each conjoint task presents two experimentally generated policy profiles. Respondents select the proposal they prefer, producing a binary choice outcome for each profile. The primary outcome measures the relative attractiveness of a policy proposal within a paired comparison. This allows the study to estimate marginal means and average marginal component effects

(AMCEs) for the randomized policy attributes. The corresponding estimands are summarized in Table 1. The primary bundling effect compares the equally weighted average marginal means of the three positive-peace allocations with the all-environment baseline. Additional contrasts compare revenue compositions, peacebuilding programs, and household cost levels.

The design may also include an absolute support rating for each proposal. Unlike the forced-choice outcome, which measures relative preferences between two alternatives, this measure would capture respondents' stated willingness to support an individual policy proposal. Absolute support ratings would be particularly useful for interpreting coalition formation, as they would provide complementary evidence about whether bundling increases support among previously skeptical respondents or reduces support among existing climate-policy supporters.

3.4 Survey-order randomization and coalition analysis

In addition to the profile-level randomization, the experiment includes a respondent-level randomization of the timing of climate-policy and peacebuilding attitude questions. As illustrated in Figure 1, respondents are assigned to one of two survey-order arms. In Arm A, climate and peacebuilding attitudes are measured before the conjoint experiment. In Arm B, the same attitudes are measured after the conjoint tasks. This design allows the study to assess whether eliciting policy attitudes before the conjoint influences respondents' subsequent evaluations of carbon-tax proposals. More importantly for the coalition analysis, Arm A provides pre-treatment measures of respondents' orientations toward climate policy and peacebuilding. As specified in Table 1, RQ2 distinguishes four baseline constituencies: respondents supportive of both objectives, respondents supportive of climate policy but skeptical of peacebuilding, respondents skeptical of climate policy but supportive of peacebuilding, and respondents skeptical of both. The analysis estimates the bundling effect separately within these constituencies and compares the resulting effects across groups.

This approach distinguishes whether introducing peacebuilding attracts individuals who would otherwise be reluctant to support climate policy from whether it alienates respondents who already favor environmental objectives. The primary constituency analysis uses Arm A because attitudes measured after exposure to the conjoint cannot be treated as pre-treatment characteristics. Arm B instead contributes to the assessment of survey-order effects and the overall analysis of randomized policy attributes.

4 Power Analysis and Sample-Size Planning

I conducted a Monte Carlo power analysis to assess the sample size needed to estimate the effects of peace earmarking and the heterogeneity of those effects across climate and peace constituencies. The literature-based assumptions and theoretical motivations are documented in the companion `benchmark_review.md`, while the implemented simulations, numerical results, and reproducible code are in `power_analysis.Rmd`. The benchmarks draw on Bergquist, Mildenberger, and Stokes (2020), Beiser-McGrath and Bernauer (2019), Baranzini and Carattini (2017), and Mohammadzadeh Valencia and coauthors (2024). Their research informs plausible effect magnitudes but does not provide direct estimates of the effects of climate–peace earmarking in Colombia. In particular, adding a social program to a climate-policy package is not equivalent to reallocating a fixed pool of environmental-tax revenue toward peacebuilding. The assumed effect sizes should

therefore be understood as planning inputs rather than predictions.

The simulations reproduce the six paired-choice tasks, the two survey-order arms, the conditional assignment of peacebuilding programs, and the full set of conjoint attributes. Choices are generated using a joint logistic random-utility model with respondent-level preference heterogeneity. The resulting data are analyzed with linear probability models and respondent-clustered standard errors, as planned for the conjoint. For each scenario and sample size, I estimate statistical power as the proportion of 500 Monte Carlo replications rejecting a two-sided null hypothesis at the 5% significance level. I report Monte Carlo standard errors and examine sample sizes of 500, 1,000, 1,500, 2,000, 2,500, 3,000, and 4,000 completed respondents. I also calculate Holm-adjusted power separately for the family of six between-constituency contrasts in RQ2 and the three revenue-allocation contrasts in RQ3a. The confirmatory treatment of RQ1 and the focal RQ2 comparison as two co-primary tests will be specified in the final multiple-testing plan.

My two central comparisons are the average effect of introducing positive peace earmarking (RQ1) and the difference in that effect between pro-climate/peace-skeptical and climate-skeptical/pro-peace respondents (RQ2). Under the baseline assumptions, a sample of 2,000 completed surveys provides 97.2% estimated power for RQ1, with a simulated average effect of approximately -2.95 pp, and 94.0% *unadjusted* power for the focal RQ2 difference. These estimates are conditional on the assumed treatment effects and constituency composition. They do not imply that all individual constituency effects or all comparisons between groups can be detected with comparable precision. Because RQ2 uses only Arm A, its effective respondent sample is approximately half the overall sample. Increasing the sample from 2,000 to 3,000 completed respondents improves precision for several secondary allocation comparisons. Under the baseline assumptions, unadjusted power for the 50%-versus-25% peace contrast rises from 59.2% to 74.4%, and power for the 75%-versus-50% contrast rises from 57.0% to 79.0%. I therefore consider 3,000 completed and usable surveys a reasonable preliminary recruitment target, subject to cost and feasibility, rather than a sample size strictly required to detect the two focal comparisons under the baseline model.

The sensitivity analyses qualify this recommendation. At $N = 2{,}000$, estimated power for RQ1 falls from 97.2% in the baseline to 23.0% when the assumed average effect is approximately -1 pp. With constituency shares of 50%, 20%, 20%, and 10%, it falls to 26.4%; in that scenario, the composition change also shifts the implied average treatment effect to approximately -1.04 pp, rather than merely reducing subgroup precision. The minimum-detectable-effect exercise at $N = 2{,}000$ reports approximate 80%-power thresholds on a discrete grid that scales all revenue-allocation effects together. Those thresholds are not independently optimized for each contrast, and the plotted tests are unadjusted.

Table 1: Mapping Research Questions to Experimental Design and Econometric Estimands

Research question	Attributes and levels	Econometric estimand	Substantive interpretation
RQ1. Does peace earmarking increase climate-policy support?	**Revenue allocation:** 100% environment / 0% peace 75% environment / 25% peace 50% environment / 50% peace 25% environment / 75% peace	**Average bundling effect:** $\frac{MM_{25} + MM_{50} + MM_{75}}{3} - MM_0$ Equal-weighted average of the three positive-peace allocations relative to the all-environment baseline.	Estimates whether introducing peacebuilding into the allocation of carbon-tax revenues increases or decreases policy support, on average, relative to exclusively environmental earmarking. The effect averages over the specified positive peace shares and eligible peacebuilding programs.
RQ2. Does bundling expand, substitute, or contract the coalition?	**Conjoint attribute:** Revenue allocation (as in RQ1) **Baseline constituencies (g):** Pro-climate / pro-peace Pro-climate / peace-skeptical Climate-skeptical / pro-peace Skeptical of both *Measured before the conjoint in survey-order Arm A.*	**Heterogeneous bundling effects:** $\tau_B(g) = \frac{MM_{25,g} + MM_{50,g} + MM_{75,g}}{3} - MM_{0,g}$ **Between-group contrasts:** $\Delta_{g,h} = \tau_B(g) - \tau_B(h)$ Compare bundling effects across pre-treatment constituencies.	Identifies which constituencies become more or less supportive when peacebuilding is introduced. Positive effects among initially climate-skeptical groups suggest potential recruitment; negative effects among existing climate supporters suggest possible alienation. The pattern of gains and losses informs whether bundling may expand, substitute, or contract the coalition of support.
RQ3a. How does the allocation of carbon-tax revenue between environmental and peacebuilding objectives shape public support?	**Positive peace allocations:** 75% environment / 25% peace 50% environment / 50% peace 25% environment / 75% peace *Comparisons are among bundled policies only.*	**Pairwise composition effects:** $\tau_{50-25} = MM_{50} - MM_{25}$ $\tau_{75-25} = MM_{75} - MM_{25}$ $\tau_{75-50} = MM_{75} - MM_{50}$ Equivalently, differences between appropriately estimated AMCEs.	Tests whether support changes as bundled policies allocate relatively more revenue to peacebuilding and less to environmental objectives. Distinguishes preferences for environment-prioritizing, balanced, and peace-prioritizing bundles without requiring a comparison against the no-bundling baseline.
RQ3b. Does the type of peacebuilding expenditure condition the political appeal of climate–peace policy bundles?	**Peacebuilding program:** Victim reparations Ex-combatant reintegration Illicit-crop substitution Rural infrastructure *Conditional: randomized only when the peace share is positive.*	**Pairwise conditional effects:** $\tau_{p-q} = \beta_p - \beta_q$ Compare AMCEs across peacebuilding programs.	Examines whether the political appeal of climate–peace bundling depends on the specific peacebuilding objective being funded. Different programs may attract new constituencies or alienate existing climate-policy supporters, helping explain why some bundles expand support while others generate opposition. Distinguishes the role of peacebuilding program type from the overall allocation of revenues between climate and peace objectives.
RQ3c. How do household costs affect support for carbon-tax policies?	**Monthly household cost:** COP 5,000 COP 15,000 COP 30,000 COP 60,000	**Pairwise cost effects:** $\tau_{c-c'} = \beta_c - \beta_{c'}$ Compare AMCEs across household cost levels, using COP 5,000 as the reference category.	Estimates how increasing the household cost of a carbon-tax proposal changes public support, holding the distribution of other randomized policy attributes constant.

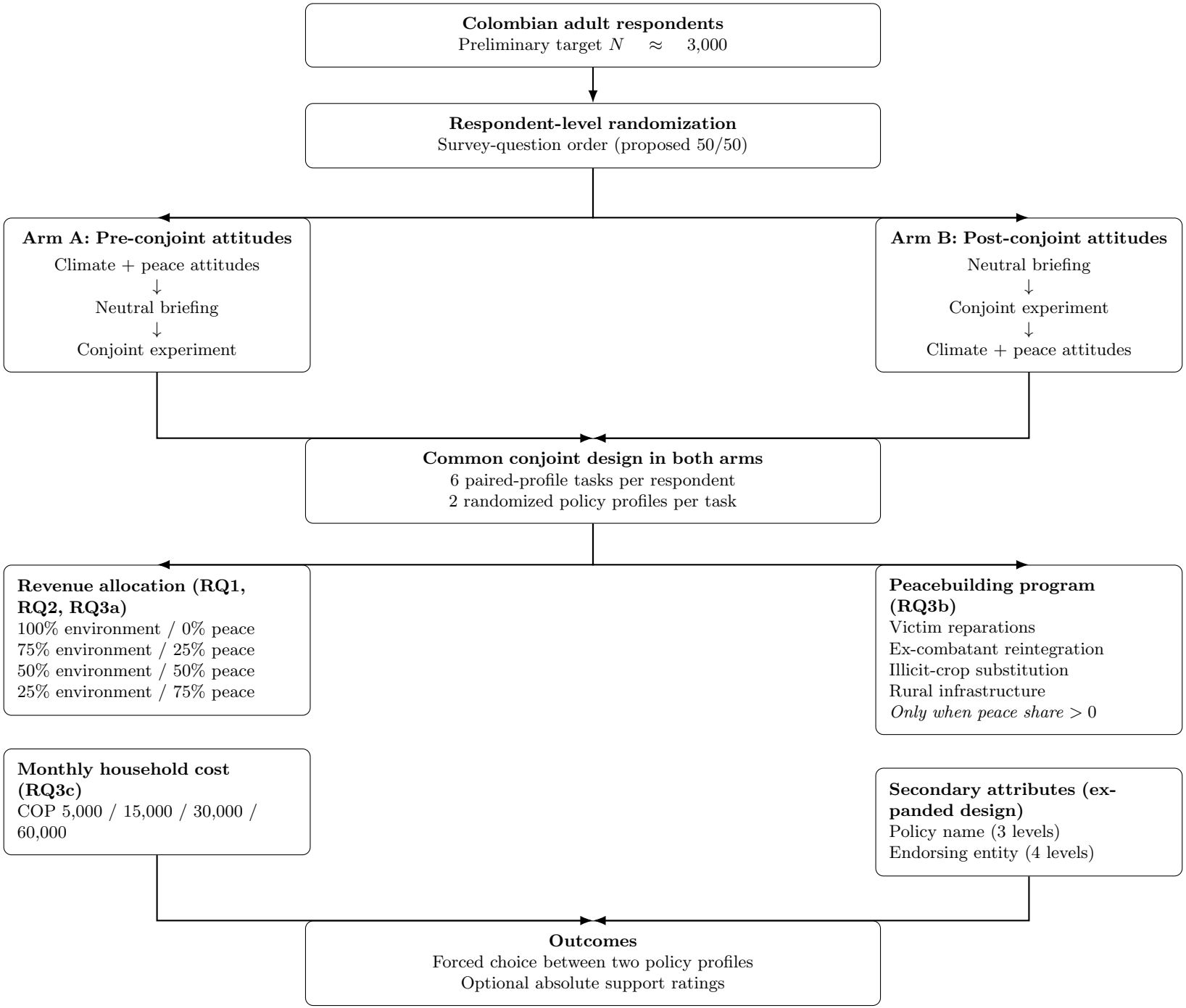


Figure 1: Hierarchical randomization and conjoint experiment design.